Software Requirements Specification
Software Engineering

Team #14, Troitsky Toolbox
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Table 1: Revision History

Date	**Developer(s)**	**Change**
10/02/2026	Pedro	SRS template (Meyer) adopted; section files created.
10/07/2026	Pedro	G.1 drafted from problem statement.
10/08/2026	Jack	P.6 risks and mitigation analysis added.
10/11/2026	Jack	P.6 expanded.
10/13/2026	Pedro	Formatting aligned with other team documents.

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer. Changes made to the template are listed in Table 4 (Appendix A.2).

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G Goals

Comment: *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

G.1 Context and overall objective

Comment: *High-level view of the project: organizational context and reason for building a system.*

The McMaster Troitsky club is a student-run organization that registers and supports 4-6 teams of students as they take on the task of modelling, analyzing and manufacturing a bridge out of popsicle sticks, white glue and dental floss. While the club has been successful in the past, results often vary in performance both between years and teams, due to the lack of centralized information and technical support for members. As teams progress through several years of competition, undergoing material testing, bridge design and analysis, and even experimenting with more innovative concepts, skill and knowledge within the club often consolidates within more experienced members or single teams, ensuring that when those members graduate, information is lost to the club. This project seeks to create a centralized information repository and technical support system for the McMaster Troitsky club, ensuring information transfer between years and teams, and facilitating support and onboarding for new members.

G.2 Current situation

Currently, teams operate in a highly individualistic manner, with the greater McMaster Troitsky club covering the administrative details regarding competition registration and providing guidance and feedback when possible. Teams store information through Microsoft Teams, Google Drive and emails between members, with many teams resulting in having a single member in charge of different competition components. Currently, the club as a whole does not have a centralized system in place to store past designs, bridge models or technical details, only having more adiminstrative/executive documents stored. Even tutorials that are consistently useful for teams do not have a consistent access point and therefore have become unreliable as proper support for the club. To address these challenges, for the past 2 years the club has been pushing for more integration between teams, generalizing the coaching positions, providing larger-scale support and creating a stronger administrative presence through timelines and events, but still lacks a service to centralize and make this data available.

G.3 Expected benefits

Comment: *New processes, or improvement to existing processes, made possible by the project's results.* Comment: *This section should not be empty!*

G.4 Functionality Overview

High Level functional requirements for the toolbox are defined below:

- **Data Storage and Retrival.** Troitsky Toolbox must store past designs, materials testing informat, and relevant development information. Troitsky toolbox must be able to return relevant data quickly based on user searches. The system should allow for the addition of new information, streamilining the process of growing the repository.
- **Key Model Analysis Features and Bridge Predictions.** Provide a platform for teams to upload their bridge models through a standardized format, allowing the user to view and analyze the model, highlighting key metrics and predictions about the bridge's performance. This should decrease the barrier to entry to bridge analysis, allowing all members of a team to get involved in the analysis process, even without formal modelling knowledge. This will be integrated with the material testing support to provide more accurate bridge analysis from the stored material properties.
- **Materials Testing Support and Facilitated Result Analysis** As the McMaster Troitsky club begins to standardize knowledge collection, material testing data will be collected and stored in a standardized format. The service will streamline this process, allowing testing results to be easily uploaded and analyzed against past results, along with collecting supplementary testing information for future years.
- **Simplified and Comprehensive Knowledge Transfer** Support documentation from the McMaster Troitsky club through exit documents and other knowledge transfer methods, ensuring that information and designs are consolidated and accessible to future members. Specialized member roles, such as executive positions, will also be able to keep a record of their role responsibilities and key factors, to further improve the club organization.

G.5 High-level usage scenarios

Comment: *Fundamental usage paths through the system.*

G.6 Limitations and exclusions

Below is a list of limitations for the system, items that are currently out of scope for development:

- The system is not looking to replace the more complex modelling capabilities of external/industry standard modelling software. Unlike the current apps being used by teams, such as AutoCAD Inventor, SAP2000 and SolidWorks, the modelling component within Troitsky Toolbox will be intentionally simplified, which will leave many of the more advanced features out of scope.
- The system will not support user access for past members. The service will only be accessible by current members, with only the public-facing landing page and gallery available to others.
- The system will not replace all forms of communication between members. Given the accepted use of other platforms for communication and the ease of use in utilizing an existing app for communication, peer-to-peer or individual team channels will not be supported in the toolbox.
- The system will not replace registration into the club; initial applications into specific roles will remain external. Payment support will be excluded and done externally from the service with registration.

G.7 - Stakeholder and Requirements Sources

Table 2: Stakeholder Personas

Stakeholder	Persona	Category
McMaster Troitsky Executives	Mohammed Lee	Direct
General McMaster Troitsky Members	Tenun Grigoryan	Direct
Future Troitsky Members	Sui Smith	Direct
McMaster CAS Students	Kyle Levi	Direct
MES - McMaster Engineering Society	Mia Tanaka	Direct
Concordia University Troitsky Organization	Persona	Indirect
McMaster University	Persona	Indirect

G.7.1 - Direct Stakeholders

McMaster Troitsky Executives
This toolbox web service was requested by the Troitsky Executives; they have noticed issues with information carryover during executive transitions, as well as information and technical skill consolidation within the most senior teams. They would like a centralized application that can address these issues.

Persona: Mohammed Lee
Mohammed is a VP on the Troitsky Exec team. He has competed in Troitsky three times: twice as a team member and once as a coach. Through his experience on the team, he has developed a wealth of knowledge in both the

administrative and technical sides of the club. He is graduating next year and is worried about the transition process for the new VP taking his place.

General Troitsky Members
General members will be the main users and the primary focus of our project. These users will both support and expand the data the system provides, and will be the main beneficiaries of the system.

Persona: Tenun Grigoryan
Tenun is a second-year member of Troitsky. He has competed on a team once before and is hoping to compete again this year. In his first year, there was a steep learning curve in the CAD design aspect of the competition, and he had to rely heavily on senior Troitsky members for tutorials and help. He hopes that this year, more resources will be provided to junior club members to decrease the barriers to competing.

Future Troitsky Members
Future Troitsky members are students coming to McMaster who will carry the torch of the Troitsky Club. This toolbox will lower the barrier to entry for new teams wanting to join Troitsky through both access to previous teams' work and documentation, as well as scripted load testing and weak-point analysis.

Persona: Sui Smith
Sui is a high school senior who has just been accepted into the McMaster Civil Engineering Program. She is part of student parliament at her school and competes in robotics competitions as one of her extracurriculars. She would love to continue competing on a technical team that aligns with her program. It goes without saying that she would prioritize joining a club that provides good resources for its introductory members.

McMaster CAS Students
Troitsky will need a software development team to maintain and extend the toolbox. This team will provide new extracurricular activities for CAS students at McMaster, allowing them to develop essential software engineering skills that will give them a leg up in school and work.

Persona: Kyle Levi
Kyle is a third-year CS student. He wants to work in large-scale system design and management when he graduates; he would like to join a technical team that focuses on maintaining and expanding a software system.

MES - McMaster Engineering Society
The MES organizes engineering clubs and seeks to benefit from the legitimization of the McMaster Troitsky Club through a centralized information platform. The MES also provides funding for the McMaster Troitsky Club, so the hosting of services and associated costs may be dependent on the given budget.

Persona: Mia Tanaka
Mia is the treasurer for the MES. She would like to ensure that the club funding she is approving is going to clubs that are investing in their long-term sustainability and development.

G.7.2 - Indirect Stakeholders

Concordia University Troitsky Organization
As Concordia University hosts the Troitsky competition, it establishes the rules and regulations of the competition and therefore influences what is prioritized by the direct stakeholders. This would influence the design of the system and which features would benefit teams the most.

McMaster University
The university itself is an indirect stakeholder, as it benefits from the success of the McMaster Troitsky Club and the success of teams during the Troitsky competition through an improved reputation as a school that produces capable students.

E Environment

Comment: *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

E.1 Glossary

Comment: *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

E.2 Components

The system will interact with the following external components:

- **Official Troitsky Organization - Concordia:** As the Troitsky club is generally officiated and organized fully by Concordia University's Troitsky organization, much of what is valued in the product depends on what is valued in the competition itself. Between years, rules are often revised, and different competition components are added, removed or modified. As is the case, the product will be heavily dependent on the organization and must allow for compliance and rule checking in line with the current competition standards.
- **CAD Software Modelling Tools:** This is used by teams to create their STEP files, which will serve as the input for the Model Analysis.

- **Static Model Analysis APIs:** For the modelling component of the product, external APIs will be used to perform the static analysis of the inputted STEP files. This will be used to calculate/create the output predictions and model data.
- **Hosting and Data Storage Platform:** The product will require a platform or server to host the website and the shared database. This is where the repository of information, uploaded files and backups will be stored.
- **Materials Testing Department Setup:** As the product seeks to support the material testing department, the app will need to work around the data formats and the processing the team prefers.
- **Club Sponsors and University Funding:** The MES and external sponsors dictate the funding that the club receives yearly. This funding may dictate the future upkeep of the product, being required for the hosting and data storage platform and potentially the external modelling API usage.

E.3 Constraints

Troitsky Toolbox Constraints

- **Competition Rules:** The model compliance checker and other support services within Troitsky Toolbox must comply with the rules set out for the competition. Outdated guides must be labelled to ensure users understand they do not fit with current competition rules.
- **Member Data and Privacy:** As the service seeks to compile team and member information, member privacy and sharing permissions dictate what can be collected. During the current competition year, individual team files must be left separate, with only executives and coaches being able to view all files. After the competition is complete, exit documentation collected will determine what individuals wish to have stored for future years.
- **Club Data Format Compatibility :** The system must accommodate the file formats that the club utilizes, being constrained to a generalized file format when multiple are available. In the case of modelling files, STEP (.stp) files must be used as input as they are available across utilized services. The material testing data collection must accept .csv files, working with the data exporting the club utilizes.

Operational Constraints

- **Budget:** The funding set out by the MES, along with the yearly club sponsorships, will determine the budget and therefore constrain the services and upkeep of the product. Troitsky Toolbox must operate within

its share of the overall club funds, dictating the storage capabilities and API usage of the service.

- **Troitsky Software Subteam:** Maintenance year to year after this year will be limited by the number of software subteam members the club recruits. As the subteam is made up of student volunteers, the experience and drive of the members will also determine the level of development of the system.

E.4 Assumptions

The following properties of the environment are assumed.

- **Competition Format Stability:** Despite the product being adaptable to changes within rules and smaller operational changes, the overall format of the Troitsky competition is assumed to remain stable across years. While some forms of bridges, materials or scoring are revised, the product serves the standard competition format of popsicle stick, white glue and floss bridges tested for ultimate point load at the centre. The rule set may change across years, but competition rulebooks are assumed to be published at the start of each competition year, allowing for integration into the rule compliance checker once yearly, with only minor modifications necessary.

- **User Access and Device Availability:** As the product will be available through a website, there is an assumption that members will both have access to a suitable device (desktop or laptop) and a standard internet browser. There is also an assumption that all users will have a working internet connection when accessing the service.

- **McMaster Troitsky Club Organizational Structure:** The role-based access control of the system will be designed around the current club structure, specifically around the roles in the organizational body of the club (general members, team captains, coaches, executives). It is assumed that this structure will not change frequently and that the access of each role will remain constant across competition years.

- **Continued Club Operation:** There is an assumption that the Troitsky competition, both as a whole and as McMaster's individual club, will continue to operate. This includes the assumption that teams will continue to register and that the software subteam will continue to have registered members.

- **Model File Conventions:** Bridge models are assumed to be exported in the STEP file format. This export will be in line with a standard export format outlined by the club, allowing for some consistency in how bridge members and units are represented across files. In line with P.6, these file conventions are linked to the most complex and high-risk component

of the product, so an agreed-upon standard simplifies the scope of the service, serving as a mitigation strategy for the risk.

- **Material Behaviour:** There is an assumption of some consistency in material properties while the club purchases materials from the same supplier. As the model analysis predictions may rely on the measured material properties of popsicle stick and glue slabs, constant material properties are a required assumption to create any form of performance prediction.

E.5 Effects

Comment: *Elements and properties of the environment that the system will affect.*

- Centralization of development information - the wealth of development information (e.g. past designs, material testing information) will be in one central location for future members to access, which will facilitate onboarding.
- Model analysis - team members will be able to submit their models and recieve key metrics and predictions regarding that model through an automatic analysis.
- Facilitated comparison - team members will be able to easily compare modelling results for different models. This will facilitate development because team members will be able to start from the best and abandon changes that caused the model to degrade.
- Simplified knowledge transfer - veteran team members will be able to share their vast knowledge with new members more easily. This is critical, especially for outgoing team members.
- Creation of a permanant software sub-team - this project will be the genesis of a permanant sofware subteam of the club. Furthermore, brief but top-tier resources alongside the software suitewill be developed to smooth the onboarding of new software subteam members.

E.6 Invariants

Comment: *Properties of the environment that the system's operation must preserve.*

- The knowledge entered into the system, including development knowledge, scientific and engineering methods, and veteran wisdom, shall be fully preserved.
- Any model entered into the system, whether developed before or after the system's released, shall be fully preserved and not corrected at all to preserve development history as faithfully as possible for easy comparison.

- The rules of the competition that are relevant to the system shall be fully preserved and not be mutated at all. This is because modelling will be based of the rules as they are, and any deviation from the rules will result in inferior models.

S System

Comment: *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

S.1 Components

Comment: *Overall structure expressed by the list of major software and, if applicable, hardware parts.*

The Troitsky Toolbox is a web-based application structured around a central broker. Each tool, as well as the shared data storage, is a service that registers with the broker. All communication between the user interface and the services, and between the services themselves, passes through the broker, so that no service depends directly on another. The system relies on the following software components:

- **Web Server:** The component in charge of hosting the Toolbox Web App and serving it to members' browsers. It receives all incoming requests (`Handle Request`), authenticates members at login through the Access Control System, and forwards requests for club services to the Toolbox Broker. It also serves the public club landing page, which requires no authentication.

- **Access Control System:** The component in charge of managing member accounts and their roles (`Manage Members`), such as general member, executive, and software subteam maintainer. It also records which team, if any, each member belongs to. It authenticates users (`Authenticate`) and decides which services and data each member may access based on their role and team (`Authorize`), ensuring that club services and stored data are limited to authorized members, and that a team's in-progress work is limited to that team.

- **Toolbox Broker:** The component in charge of coordinating all communication within the system. Services register with the broker (`Register Service`), advertising the operations they provide. The broker receives each request (`Route Request`), verifies it with the Access Control System (`Authorize`), and forwards it to the service that provides the requested operation. New tools can be added to the Toolbox by registering them with the broker, without modifying the broker or any existing service.

- **Knowledge Repository:** The service in charge of the centralized, persistent store of all club information (`Manage Data`), including documents, current and past bridge models, material test results, competition rules, and handoff documents. Other services store and retrieve their data by sending requests through the broker, which allows information produced by one tool to be found and reused by members and by other tools. It also allows members to search, filter, and contribute content (`Search Content`).

- **Model Analysis System:** The service in charge of importing bridge models in a standardized 3D format (`Upload Model`), rendering an interactive visualization of the model, and extracting the geometric and structural data of the bridge (`Model Data`). It works with two kinds of bridge models, both stored in the Knowledge Repository:

  - **Current team models:** The models a team is developing for the upcoming competition. These are accessible only to the members of that team, so that teams can work on their designs independently.
  - **Past designs:** Models of bridges from previous years, which any member can load into the model viewer to inspect and learn from, but cannot modify. At the end of each competition season, a team's current models become past designs.

  Its analysis capabilities are provided by two subsystems, which extend its functionality while remaining separate from it and from each other:

  - **Performance Analysis System:** Computes key metrics and predictions about a bridge's expected performance (`Analyze Performance`), using the extracted model data and the material properties provided by the Material Testing Manager, both requested through the broker.
  - **Rule Compliance System:** Assesses a bridge model against the current Troitsky competition rules (`Check Compliance`), using the extracted model data requested through the broker. It maintains the set of rules being checked (`Manage Rules`) so that they can be updated as the competition rules change from year to year.

- **Material Testing Manager:** The service in charge of facilitating the storage and access of material testing results in a standardized format (`Manage Test Results`), and analyzing new results against those collected in previous years. This service also derives material-specific data from accepted test results (`Material Properties`), which is provided to the Performance Analysis System through the broker.

- **Knowledge Transfer Manager:** The service in charge of collecting and organizing exit documents, role records, and onboarding material (`Manage Handoff Documents`), so that the knowledge held by graduating members and executive positions is preserved for future members.

In addition to these components, the system will interact with external components, such as a CAD/geometry library used by the Model Analysis System to read STEP files (see E.2 for details).

Club members interact with the system through the **Toolbox Web App**, a browser-based interface served by the Web Server, which presents each of the tools above. Through it, members can browse and contribute to the repository, upload and analyze bridge models, record material tests, and complete handoff documents. Members of the software subteam use the same interface, with elevated privileges granted by the Access Control System, to maintain accounts, competition rules, and stored content.

Figure 1 shows the components of the system and the interfaces through which they interact.

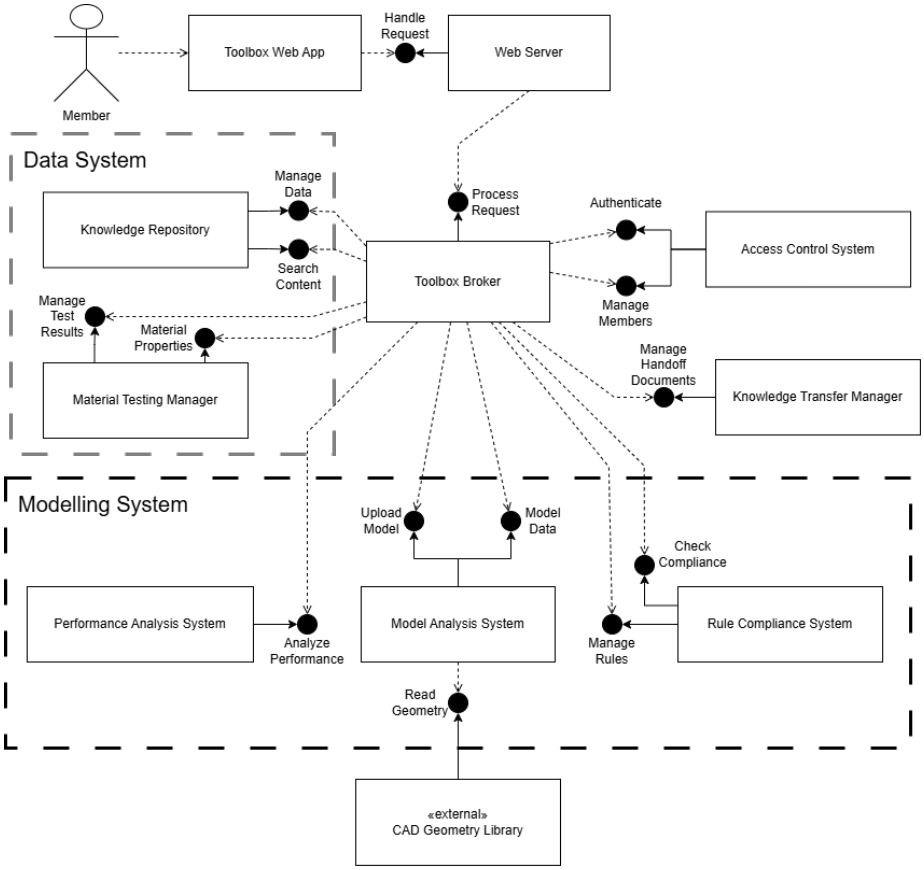


Figure 1: Component diagram of the Troitsky Toolbox. All communication between services passes through the Toolbox Broker.

S.2 Functionality

Comment: *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).*

This section describes the behaviour of each component identified in S.1. Each subsection lists the functional and non-functional requirements of one component. Requirements are labelled with a prefix identifying their component, followed by a number: for example, FR-BR-1 is the first functional requirement of the Toolbox Broker, and NFR-BR-1 its first non-functional requirement. The prefixes are WS (Web Server), AC (Access Control System), BR (Toolbox Broker), KR (Knowledge Repository), MA (Model Analysis System), PA (Performance Analysis System), RC (Rule Compliance System), MT (Material Testing Manager), KT (Knowledge Transfer Manager), and SYS (system-wide requirements). The symbolic constants used by the non-functional requirements are defined in Table 3 (Appendix A.1).

S.2.1 Web Server

Functional Requirements

1. **Public landing page:** Web Server shall serve a public club landing page to any visitor, without requiring authentication. (FR-WS-1)

2. **Login and logout:** Web Server shall allow members to log in by submitting their credentials to the Access Control System, and to log out, ending their session. (FR-WS-2)

3. **Request forwarding:** Web Server shall forward every request for a club service from an authenticated member to the Toolbox Broker, along with the identity of the member making the request. (FR-WS-3)

Non-Functional Requirements

1. **Serve web app:** Web Server shall serve the Toolbox Web App to members' browsers. (NFR-WS-1)

2. **Encrypted communication:** Web Server shall only communicate with browsers over an encrypted connection (HTTPS). (NFR-WS-2)

3. **Session expiry:** Web Server shall end a member's session after SESSION_TIMEOUT (30 minutes) of inactivity, so that an unattended browser does not leave club data accessible. (NFR-WS-3)

S.2.2 Access Control System

Functional Requirements

1. **Authentication:** Access Control System shall verify the credentials submitted by a member, and shall reject login attempts with invalid credentials or for deactivated accounts. (FR-AC-1)

2. **Roles:** Access Control System shall assign each account exactly one role: general member, team member, team captain, club executive, or software subteam maintainer. (FR-AC-2)

3. **Authorization:** Access Control System shall determine, for each request it is asked to verify, whether the current role and team of the requesting member permit the requested operation. (FR-AC-3)

4. **Member management:** Access Control System shall allow executives and maintainers to create accounts, deactivate accounts, and change the role of an account, so that membership can be updated as members join, are promoted or demoted, graduate, or otherwise leave. (FR-AC-4)

5. **Team membership:** Access Control System shall allow executives and maintainers to assign each account to at most one team for the current competition year, and to change or remove that assignment. (FR-AC-5)

6. **Team data isolation:** Access Control System shall deny any member who is not on that team, including former members, access to that team's current bridge models and analysis results. (FR-AC-6)

Non-Functional Requirements

1. **Credential security:** Access Control System shall store passwords only as salted hashes produced by an industry-standard algorithm, and shall never store passwords in plain text. (NFR-AC-1)

2. **Effective role alterations:** Access Control System shall ensure that if a user's role is altered, the change shall be effective by the user's next request to the system. (NFR-AC-2)

S.2.3 Toolbox Broker

Functional Requirements

1. **Service registration:** Toolbox Broker shall allow services to register at system startup, recording the operations that each registered service provides. (FR-BR-1)

2. **Request authorization:** Toolbox Broker shall verify every request with the Access Control System before forwarding it, and shall reject requests that are not authorized. (FR-BR-2)

3. **Request routing:** Toolbox Broker shall forward each authorized request to the registered service that provides the requested operation, and shall return that service's response to the requester. (FR-BR-3)

4. **Unavailable services:** Toolbox Broker shall return an error message to the requester when the requested operation is not provided by any registered service, or when the service providing it does not respond. (FR-BR-4)

Non-Functional Requirements

1. **Extensibility:** Toolbox Broker shall allow a new service to be added to the system by registering it, without modifying the source code of the broker or of any existing service. (NFR-BR-1)

2. **Routing overhead:** Toolbox Broker shall add no more than MAX_BROKER_OVERHEAD (100 ms) to the response time of any request it routes. (NFR-BR-2)

3. **Request logging:** Toolbox Broker shall log each request it handles, including the requesting member, the requested operation, the time, and whether the request succeeded. (NFR-BR-3)

S.2.4 Knowledge Repository

Functional Requirements

1. **Data storage:** Knowledge Repository shall store and return the data of every other service, including documents, bridge models, material test results, competition rules, and handoff documents. (FR-KR-1)

2. **Content upload:** Knowledge Repository shall allow members to upload files of any common type, along with descriptive metadata including title, category, year, team, and tags. (FR-KR-2)

3. **Search and filter:** Knowledge Repository shall allow members to search stored content by keyword across titles and metadata, and to filter results by category, year, team, and tag. (FR-KR-3)

4. **View and download:** Knowledge Repository shall allow members to preview supported file types in the browser, and to download any stored file. (FR-KR-4)

5. **Content management:** Knowledge Repository shall allow the uploader of an item, and executives, to edit its metadata or archive it. Archived items shall remain retrievable, but shall be excluded from default search results. (FR-KR-5)

Non-Functional Requirements

1. **Search responsiveness:** Knowledge Repository shall return search results within MAX_SEARCH_TIME (2 seconds) for a repository containing up to MAX_REPOSITORY_ITEMS (10,000 items). (NFR-KR-1)

2. **Data durability:** Knowledge Repository shall be backed up at least every BACKUP_INTERVAL (7 days), so that no more than BACKUP_INTERVAL of stored data can be lost in the event of a failure. (NFR-KR-2)

S.2.5 Model Analysis System

Functional Requirements

1. **File validation:** Model Analysis System shall reject files that are not valid STEP files, or that contain no readable geometry, and shall display an error message describing the problem. (FR-MA-1)
2. **3D visualization:** Model Analysis System shall render an interactive 3D view of an uploaded model, which the member can rotate, zoom, and pan. (FR-MA-2)
3. **Data extraction:** Model Analysis System shall extract from each uploaded model the geometric and physical properties of the bridge required by its analysis subsystems, such as its dimensions and mass. (FR-MA-3)
4. **Current model storage:** Model Analysis System shall store each uploaded model and its extracted data in the Knowledge Repository as a current model of the uploading member's team, accessible only to members of that team, so that it can be reopened and analyzed later. (FR-MA-4)
5. **Past design viewing:** Model Analysis System shall allow any member to browse past designs, filtered by year and team, and to load any past design into the model viewer and its analysis subsystems for inspection, without being able to modify it. (FR-MA-5)
6. **Archiving current models:** Model Analysis System shall allow executives to convert all current team models into past designs at the end of a competition season, recording the competition year and team of each. (FR-MA-6)

Non-Functional Requirements

1. **Model upload:** Model Analysis System shall allow members to upload bridge models in the STEP (`.stp`/`.step`) format, which is the only model format accepted by the system. (NFR-MA-1)
2. **Load time:** Model Analysis System shall display the 3D visualization of a typical club bridge model within MAX_MODEL_LOAD_TIME (30 seconds) of upload or of being selected. (NFR-MA-2)
3. **Extraction accuracy:** Model Analysis System shall extract dimensions that match those of the source model to within DIMENSION_TOLERANCE (0.5 mm). (NFR-MA-3)

S.2.5.1 Performance Analysis System

Functional Requirements

1. **Failure load estimate:** Performance Analysis System shall estimate the failure load of a bridge model, being the maximum load the bridge can support before failure. (FR-PA-1)

2. **Load-to-mass ratio:** Performance Analysis System shall compute the estimated load-to-mass ratio of a bridge model, from its estimated failure load and its extracted mass. (FR-PA-2)

3. **Material selection:** Performance Analysis System shall use the material properties provided by the Material Testing Manager, and shall allow the member to choose which set of material data, such as a given year or material batch, to apply. (FR-PA-3)

4. **Assumption disclosure:** Performance Analysis System shall display the assumptions and limitations of each estimate alongside its results, so that members do not mistake an estimate for a guaranteed result. (FR-PA-4)

Non-Functional Requirements

1. **Estimate accuracy:** For past club bridges whose tested failure loads are recorded, Performance Analysis System shall estimate the failure load to within FAILURE_LOAD_TOLERANCE (25%) of the recorded value in competition. (NFR-PA-1)

2. **Computation time:** Performance Analysis System shall produce results for a typical club bridge model within MAX_ANALYSIS_TIME (30 seconds). (NFR-PA-2)

3. **Extensibility:** Performance Analysis System shall be structured so that additional performance metrics can be added in the future without modifying the computation of existing metrics. (NFR-PA-3)

S.2.5.2 Rule Compliance System

Functional Requirements

1. **Rule import:** Rule Compliance System shall allow executives and maintainers to import a set of competition rules for a given competition year. (FR-RC-1)

2. **Rule configuration:** Rule Compliance System shall allow executives and maintainers to create, edit, and remove individual rules, each defined by the property it constrains, such as a dimension or mass, and its allowed value or range. (FR-RC-2)

3. **Compliance check:** Rule Compliance System shall evaluate a bridge model against the active rule set, using the data extracted by the Model Analysis System, and shall report a pass or fail result for each rule. (FR-RC-3)

4. **Violation details:** For each failed rule, Rule Compliance System shall report the violated rule, the measured value, and the allowed value or range. (FR-RC-4)

5. **Manual checks:** Rule Compliance System shall list the rules that cannot be verified from the model, such as restrictions on materials or construction methods, as manual checks for the team to confirm. (FR-RC-5)

Non-Functional Requirements

1. **Check correctness:** Rule Compliance System shall report no false passes on a test suite of models with known rule violations. (NFR-RC-1)
2. **Rule updatability:** Rule Compliance System shall allow an executive without programming knowledge to configure the rule set for a new competition year without modifying source code. (NFR-RC-2)

S.2.6 Material Testing Manager

Functional Requirements

1. **Result entry:** Material Testing Manager shall allow members to record individual test results through a form, including the test type, material, specimen dimensions, applied load, date, and tester. (FR-MT-1)
2. **Input validation:** Material Testing Manager shall reject test results that are missing required fields, or that contain physically impossible values such as negative dimensions or loads, and shall identify the invalid rows or fields. (FR-MT-2)
3. **Historical comparison:** Material Testing Manager shall compare new test results against results from previous years, displaying summary statistics (mean and standard deviation) and plots. (FR-MT-3)
4. **Material properties:** Material Testing Manager shall compute aggregated material properties from accepted test results, and shall provide them to the Performance Analysis System. (FR-MT-4)
5. **Result review:** Material Testing Manager shall allow executives to flag test results as invalid, excluding them from aggregated material properties without deleting them. (FR-MT-5)

Non-Functional Requirements

1. **Result import:** Material Testing Manager shall allow members to import material test results from CSV and Excel (`.xlsx`) files that follow a standardized template. (NFR-MT-1)
2. **Traceability:** Material Testing Manager shall ensure that every aggregated material property can be traced to the individual test results it was computed from. (NFR-MT-2)
3. **Export:** Material Testing Manager shall allow stored test results to be exported in CSV format, so that the data remains usable outside the system. (NFR-MT-3)

S.2.7 Knowledge Transfer Manager

Functional Requirements

1. **Exit document templates:** Knowledge Transfer Manager shall provide an exit document template for each executive position and for team leads. (FR-KT-1)

2. **Exit document submission:** Knowledge Transfer Manager shall allow members to complete, save as drafts, and submit exit documents. (FR-KT-2)

3. **Role records:** Knowledge Transfer Manager shall maintain a record for each executive position, describing its responsibilities, key contacts, and yearly timeline, which the current holder of the position can update. (FR-KT-3)

4. **Publication:** Knowledge Transfer Manager shall store submitted exit documents and role records in the Knowledge Repository, where they can be searched like other content. (FR-KT-4)

5. **Outstanding documents:** Knowledge Transfer Manager shall allow executives to view which executives and team leads have not yet submitted their exit documents. (FR-KT-5)

Non-Functional Requirements

1. **Findability:** Knowledge Transfer Manager shall allow a new member to locate the role record for a given executive position within MAX_ROLE_SEARCH_TIME (2 minutes), as measured in usability testing. (NFR-KT-1)

2. **Longevity:** Knowledge Transfer Manager shall allow exit documents and role records to be exported to PDF, so that they remain accessible independently of the system. (NFR-KT-2)

S.2.8 System-Wide Requirements

Non-Functional Requirements

1. **Browser support:** The Toolbox Web App shall be fully functional on the current versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari on Windows, Macintosh, or Linux personal computers. (NFR-SYS-1)

2. **Capacity:** The system shall support at least MIN_CONCURRENT_USERS (50 members) using it concurrently without exceeding the response times specified in this section. (NFR-SYS-2)

3. **Service independence:** The failure of any single service shall not prevent members from using the remaining services. (NFR-SYS-3)

4. **Maintainability:** The system shall be documented well enough that a new software subteam member, with experience in C# and web development, can set up a local development environment and deploy a change within MAX_ONBOARDING_TIME (1 week). (NFR-SYS-4)

S.3 Interfaces

Comment: *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

S.4 Detailed usage scenarios

S.4.1 – Knowledge Repository Information Viewing

Use Case Name: Knowledge Repository Information Viewing
Primary Actor: User
Precondition:

- User has an account on the Troitsky Toolbox web application with general member permissions.

Trigger: A Troitsky member would like to view data stored in the Knowledge Repository.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests to open the Knowledge Repository via the Toolbox Broker.
3. User requests the information they are interested in from the Knowledge Repository via the Toolbox Broker.
4. Toolbox Broker returns the requested information.
5. User views, and optionally downloads, the information.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
- Toolbox Broker fails at some point.
    2. Request to open the Knowledge Repository fails.
    3. Search request in the Knowledge Repository fails.
    4. Toolbox Broker fails to return the requested information.
- Information is corrupted.

5. User cannot view the information.

Success Postcondition: User was able to find and access the information they were searching for.

Justification: This scenario describes a user's ability to read from the Knowledge Repository, directly aligning with FR-KR-1 and FR-KR-3.

S.4.2 – Knowledge Repository Information Addition

Use Case Name: Knowledge Repository Information Addition
Primary Actor: User
Precondition:

- User has an account on the Troitsky Toolbox web application with general member permissions.

Trigger: A Troitsky member would like to add relevant data to the Knowledge Repository.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests to open the Knowledge Repository via the Toolbox Broker.
3. User requests to upload information to the Knowledge Repository via the Toolbox Broker.
4. User specifies the data type of the information and uploads it to the Knowledge Repository.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
- Toolbox Broker fails at some point.
    2. Request to open the Knowledge Repository fails.
    3. Upload request to the Knowledge Repository fails.

Success Postcondition: The information the user wants to store is successfully uploaded to the Knowledge Repository.

Justification: This scenario describes a user's ability to write to the Knowledge Repository, directly aligning with FR-KR-2.

S.4.3 – Transition Information Addition

Use Case Name: Transition Information Addition
Primary Actor: Troitsky Executives
Precondition:

- User has executive access to the Troitsky Toolbox web application.

Trigger: A Troitsky executive wants to add to their transition documents.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests to open the Knowledge Transfer Manager.
3. Toolbox Broker confirms with the Access Control System that the user has the authority to access the Knowledge Transfer Manager.
4. Toolbox Broker directs the user to the Knowledge Transfer Manager.
5. User selects the documents they would like to fill out or edit.
6. User fills out or edits the documents and saves them.
7. Toolbox Broker saves the transition documents to the Knowledge Repository.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
    3. User is not granted access to the Knowledge Transfer Manager.
- Toolbox Broker fails.
    2. Request to open the Knowledge Transfer Manager fails.
    3. User is not directed to the Knowledge Transfer Manager.
    7. Upload request to the Knowledge Repository fails.
- User does not have executive permissions.
    2. Access Control System denies access to the Knowledge Transfer Manager.

Success Postcondition: User is able to add to and edit their transition documents, and those documents are saved in the Knowledge Repository.

Justification: This scenario describes the general flow of how transition documents are written, updated, and saved, directly aligning with FR-KT-1, FR-KT-2, FR-KT-3, and FR-KT-4.

S.4.4 – Performance Analysis

Use Case Name: Performance Analysis
Primary Actor: User
Precondition:

- User has an account on the Troitsky Toolbox web application with general member permissions.

Trigger: A Troitsky member would like to perform a performance analysis on a STEP file.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests access to the Performance Analysis System from the Toolbox Broker.
3. Toolbox Broker directs the user to the Performance Analysis System.
4. User inputs a STEP file, either newly uploaded or selected from the Knowledge Repository, into the Performance Analysis System.
5. Performance Analysis System calls an external API to estimate the load capacity and potential weak points of the bridge, and returns them as outputs.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
- Toolbox Broker fails.
    2. User is not redirected to the Performance Analysis System.
    4. Because the Toolbox Broker facilitates the transfer of data from the Knowledge Repository to the Performance Analysis System, the Performance Analysis System fails to retrieve the STEP file.
- External API calls fail.
    5. Performance Analysis System fails to return the load capacity and potential weak points.

Success Postcondition: User receives an estimated load capacity and the potential weak points for their STEP file.

Justification: This is the general path that use of the Performance Analysis System will follow, and it aligns with FR-MA-1, FR-MA-4, FR-MA-5, FR-PA-1, and FR-PA-2.

S.4.5 – Rule Compliance Verification

Use Case Name: Rule Compliance Verification
Primary Actor: User
Precondition:

- User has an account on the Troitsky Toolbox web application with general member permissions.

Trigger: A Troitsky member would like to analyze their model's compliance with the competition rules.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests access to the Rule Compliance System from the Toolbox Broker.
3. Toolbox Broker directs the user to the Rule Compliance System.
4. User inputs a STEP file, either newly uploaded or selected from the Knowledge Repository, into the Rule Compliance System.
5. Rule Compliance System calls an external API to calculate the size, weight, and other metrics of the bridge, and returns which of them fall within the competition's constraints, along with the assumptions and limitations of each estimate.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
- Toolbox Broker fails.
    2. User is not redirected to the Rule Compliance System.
    4. Because the Toolbox Broker facilitates the transfer of data from the Knowledge Repository to the Rule Compliance System, the Rule Compliance System fails to retrieve the STEP file.
- External API calls fail.
    5. Rule Compliance System fails to return the compliance results.

Success Postcondition: User learns whether their model complies with the current competition rules.

Justification: This is the general path that use of the Rule Compliance System will follow, and it aligns with FR-MA-1, FR-MA-4, FR-MA-5, FR-RC-1, and FR-RC-3.

S.4.6 – Materials Data Publication

Use Case Name: Materials Data Publication
Primary Actor: User
Precondition:

- User has an account on the Troitsky Toolbox web application with Materials Team permissions.

Trigger: A Troitsky Materials Team member would like to add or update materials data.
Main Success Scenario:

1. User logs into the web application and is successfully authorized by the Access Control System.
2. User requests access to the Material Testing Manager from the Toolbox Broker.
3. Toolbox Broker confirms with the Access Control System that the user has the authority to access the Material Testing Manager.
4. User is directed to the Material Testing Manager.
5. User inputs and saves the appropriate material property data.
6. Toolbox Broker saves the material property data to the Knowledge Repository.

Secondary Scenarios:

- Access Control System fails verification.
    1. User is unable to log into the web application.
    2. Toolbox Broker is unable to confirm the user's permissions, and cannot redirect the user to the Material Testing Manager.
- Toolbox Broker fails.
    2. User permissions cannot be confirmed, and the user is not redirected to the Material Testing Manager.
    6. Toolbox Broker cannot save the material property data to the Knowledge Repository.

Success Postcondition: Material property data is saved to the Knowledge Repository.

Justification: This scenario describes the general process the materials testing team follows to save their data, and directly corresponds with FR-MT-2.

S.5 Prioritization

Comment: *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

S.6 Verification and acceptance criteria

Comment: *Specification of the conditions under which an implementation will be deemed satisfactory.*

P Project

Comment: *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

P.1 Roles and personnel

P.1.1 Project Supervisor - Mason

The project supervisor will be responsible for coordinating internal deadlines, meetings and development timelines. The supervisor will serve as the main point of contact with administration, the supervisor and the TA, and serve as the default team lead.

P.1.2 Troitsky Club Communication Associate - Pedro

The Troitsky Club Communication Associate will be responsible for the majority of the coordination with the club organization and executive members. Surveys and interviews will be organized, distributed, and run by this member.

P.1.3 Front-end / UI Developer - Pedro, Michael

This role will be responsible for the UI of the product, both in technical displays of data and in more theme/design heavy components. They will build the web app's interface, so experience in web frameworks would be beneficial. They will be the ones primarily responsible for the usability testing in the Winter term's CD.

P.1.4 Back-end/Data Developer - Jack, Mason

This role will be responsible for the design of the data storage within the app. This will cover both the files and data the club stores (bridge files, tutorials), but also RBAC information and user/club site preferences.

P.1.5 Quality Assurance / Tester - Michael

The Quality Assurance / Tester will be responsible for maintaining a standard of quality in both code and documentation of the web app. This will involve a review of final product content and the overall design of the testing within the system.

P.1.6 Supervisor - Isabella Vescio

The supervisor for the project will provide guidance, domain expertise and general feedback for the service. As they are heavily involved in the club, they will also support interaction and integration with the club.

P.1.7 Teaching Assistant - Mohamad Mirhoseininejad

The TA will provide course-related feedback, answer administrative questions and provide feedback on expectations and course grading. They will serve as the main point of contact with course administration. General project feedback during TA meetings will also help keep the project timeline in check.

P.2 Imposed technical choices

- **Platform/delivery:** Web app for desktop browsers. Accessible through traditional browsers such as Google Chrome, Firefox, Edge, and Safari. Requires a working internet connection to utilize the service.
  Rationale: Ensures that the service is accessible by all club members. Also allows the Troitsky toolbox to act as a landing page for the McMaster club.

- **File Formats:** STEP (.stp/.step) files for loading and submitting models. CSV and Excel data formats (.csv/.xlsx) for importing material testing results.
  Rationale: Standardized format among possible external modelling software that teams currently use. Ensures compatibility with current and future model files (E.4). CSV and Excel data files are both tabular and work with current material testing data (E.2).

- **Back-end Programming languages:** C# on .NET, Python for subservices
  Rationale: LINQ availability and ease of use, and the ease of maintainability by future software subteam members; Python for accessibility to beneficial open-source libraries.

- **Testing, Linting and Coding Standard:** xUnit/bUnit + Coverlet for coverage, `dotnet format`, Standard .NET coding conventions
  Rationale: xUnit and Coverlet are widely documented; bUnit, running on top of xUnit, supports the testing of Blazor components. `dotnet`

`format` ships with the .NET SDK and can work with config files to integrate with the .NET compiler.

- **Authentication (RBAC):** ASP.NET Core Identity.
  Rationale: RBAC matches the club structure, and Identity is built into the ASP.NET Core stack.

P.3 Schedule and milestones

Comment: *List of tasks to be carried out and their scheduling.* **Comment:** *This section should not be empty!*

P.4 Tasks and deliverables

Comment: *Details of individual tasks listed under P.3 and their expected outcomes.* **Comment:** *This section should not be empty!*

P.5 Required technology elements

Comment: *External systems, hardware and software, expected to be necessary for building the system.*

- **Database server:** Microsoft SQL Server.
- **Hosting/ Deployment:** Azure App Service; Docker for containerization.
- **Version control service:** GitHub (repositories, PRs, Actions CI).
- **Delivery platform:** Desktop web browsers (Chrome, Firefox, Edge, Safari).
- **External file spec:** STEP (.stp) file format specification and compatible external modeling software.

P.6 Risks and mitigation analysis

Comment: *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

P.6.1 Risk Analysis

- **Risk: 3D Model Import and Analysis is Infeasible**
  - **Description:** Parsing STEP (.stp) files, rendering them, and extracting member geometry for the rule compliance and performance prediction tools is the most technically complex feature of the Toolbox. Available libraries may not support the required data extraction, or may not run in a web environment.

    - **Mitigation:** Prove the import and visualization pipeline in the Proof of Concept demonstration before dependent tools are built. Use an established geometry library rather than a custom parser, and agree on a standard CAD export template with the club to limit the variety of input files.
    - **Contingency:** Fall back to manual or spreadsheet entry of member geometry so the rule compliance and prediction tools remain usable without 3D import.

- **Risk: Insufficient Legacy Club Data**
    - **Description:** Past designs, material testing results and documentation are scattered across members and inconsistent in format, so the information repository may launch with little useful content.
    - **Mitigation:** Collect existing documents with the club executives early in the project and define standard upload templates for designs and material tests.
    - **Contingency:** Launch with the current season's data only, and plan for the repository to be populated over time through the upload tools.

- **Risk: Technical Issues with External Libraries**
    - **Description:** Technical issues may arise with external technologies used by tools such as bridge modelling and rule compliance, including licensing or deployment limits (for example, Autodesk APIs requiring a desktop installation).
    - **Mitigation:** Confirm licensing and deployment constraints of each library before adopting it. Team members test and validate their code before opening a pull request, and at least one other member reviews it.
    - **Contingency:** Replace the library with an open-source alternative, or reduce the affected tool's scope.

- **Risk: Unrealistic Schedule or Workload**
    - **Description:** Tasks may take longer than expected or workload may become unbalanced among team members.
    - **Mitigation:** Track tasks and milestones on the GitHub Kanban board. The Notetaker records timelines at check-in meetings, and the team revises estimates and reassigns tasks as needed.
    - **Contingency:** Because tools are largely independent, defer lower-priority tools to the software subteam's backlog rather than delaying core features.

- **Risk: User Interface is Not User-Friendly**

- **Description:** The interface may not be intuitive for Troitsky members with varying levels of technical and modelling experience.
- **Mitigation:** Run usability testing sessions with both new and experienced members (see the Usability Testing CD), gather feedback, and make iterative improvements.
- **Contingency:** Prioritize fixes to the most-used workflows and add in-app guidance for complex tools.

P.6.2 Threat Analysis

- **Threat: Unauthorized access to the repository or upload tools**
  - **Description:** A user that does not have authorized access is able to upload a file or the current repository data.
  - **Mitigation:** Every member must log in, with each request being checked against the member's role and team before it is routed and their team's models/information become visible. To limit direct external user access, idle sessions timeout, making the user account unavailable, and executives deactivate the accounts of members who leave the club.
  - **Contingency:** Utilizing the broker logs to track activity, and utilizing backups to return to a secure state.
- **Threat: Malicious, corrupted or falsified file uploads (STEP models and test data)**
  - **Description:** A member uploads a corrupted, falsified or malicious file, either by mistake or on purpose. A malformed STEP model could crash or stall the Model Analysis System. False or incorrect material test results would also taint the aggregated materials property results, skewing model results from teams that reference/utilize those results.
  - **Mitigation:** Only authenticated members can upload files, given their role and current project stage. Bridge models must be valid STEP files, and invalid files are rejected. Material test results must follow the standard CSV/Excel template and pass validation for missing or physically impossible values, with metadata filled out to club standard. Uploading user is recorded.
  - **Contingency:** Identify the user and data that was uploaded through logs and upload data. Executives can archive or exclude individual files from aggregated data results.
- **Threat: Loss of club records**
  - **Description:** Club records stored in the Knowledge Repository could be lost, either maliciously or unintentionally. Loss could come

from a failure of the hosting and data storage platform (E.2), a faulty update, or a member overwriting or removing content by mistake.

- **Mitigation:** The repository is backed up at least every BACKUP_INTERVAL (7 days). Items are archived rather than deleted, and the failure of one service does not affect the others.
- **Contingency:** Restore lost data from the most recent backup, losing at most BACKUP_INTERVAL (7 days) of contributions. Test results and handoff documents can also be exported outside the system.

P.7 Requirements process and report

McMaster Troitsky Executives
Both Pedro Azevedo and our supervisor Bella Vescio are Troitsky Executives; our requirements will be gathered through team meetings and supervisor meetings.
Important Questions to Ask:

- What information is most important to consolidate?
- When taking your position, what would you have liked to have seen within a transition document?
- How would you go about outlining the duties of a software development and support team?

The consolidation of information and ease of transition is one of the main goals in the development of this project, so we would want to hear the executives' opinions on this. Additionally, the challenge of this project is the maintainability and expandability of the code, so we must have an understanding of how future software teams are set up so that we can prepare documentation accordingly.

General McMaster Troitsky Members
A survey will be sent to general Troitsky members to elicit the overall wants and needs of members.
Questions in the survey will be:

- How many years of experience do you have in Troitsky (1-5 years)
- What is your current role?
- What is your modelling experience?
- How valuable would you find an entry-level modelling service, like implementing simple testing and weak point analysis?

- Rate (1-5) how valuable you would find having this information easily accessible regarding past McMaster Troitsky data. [**Bridge Modelling files, Joint Construction Data (specific model use), Manufacturing information, Past Team Results, Past Technical Presentation Material**]
- How does your team store data [**Microsoft Teams, Google Drive, Individuals store files, Discord, Email**]
- Rate(1-5) these features on how useful you would find them.
  - Searchable archive of past bridge designs
  - Database of material testing results (stick strength, glue, floss, etc.)
  - Bridge weight estimator from a submitted model
  - Structural efficiency calculation
  - Ultimate load and failure point prediction
  - Upload and visualize a team's bridge model in the browser
  - Exit documents/role handover records for executive positions
  - Onboarding guide and training material for new members
  - Private workspace for each team's own files and data
  - Competition rules and regulations reference
  - Club announcements and general information page
- Would you want your team files, results and progress to be available for future teams to learn from? [**Yes/No/Maybe**]
- If you could go back to being a new member, or if you're currently a new member, what tutorials would you want access to? [**Modeling Software, Manufacturing, Slab Building, Team Organization, Other**]

McMaster CAS Students
McMaster CAS students are needed for the Software Development Team; they will maintain and expand the toolbox. To ease the burden on them, we must elicit preferred code and documentation formats
Some questions we would like to ask are:

- When working with a system you did not develop yourself, rate(1-5) how helpful you find these documentation formats.
  - Markdown or other lightweight markup languages for README files.
  - Inline Documentation (Javadoc, Doxygen, XML Doc Comments, JSDoc, Docstrings)
  - Collaborative Wiki Formats, think online library documentation.
  - Modelling and Visual Formats (PlantUML, C4 Model)

- What naming conventions do you prefer?
    - PascalCase
    - IPascalCase (prefix interfaces with a capital I)
    - camelCase
    - _camelCase (prefix private and internal instance fields with an underscore)

A Appendix

A.1 Symbolic Constants

The non-functional requirements in S.2 refer to the symbolic constants listed in Table 3, rather than to fixed values, so that a value can be revised in one place as the project progresses. Each constant is given with its current value and the rationale or reference for that value. Values marked as team estimates are based on the team's judgement of the club's needs, and will be revisited once the Proof of Concept and usability testing provide measured data.

Table 3: Symbolic constants used in the non-functional requirements

Constant	**Value**	**Used in**	**Rationale / Reference**
MAX_BROKER_-OVERHEAD	100 ms	NFR-BR-2	Total response times of up to 1 s keep a user's flow of thought uninterrupted (Nielsen, *Usability Engineering*, 1993, ch. 5), so broker response is limited to only account for a portion.
SESSION_TIMEOUT	30 minutes	NFR-WS-2	OWASP Session Management Cheat Sheet: idle timeouts of 15–30 minutes are common for low-risk applications.
MAX_SEARCH_-TIME	2 seconds	NFR-KR-1	Delays between 1 s and 10 s are noticed but keep the user's attention on the task (Nielsen, 1993).
MAX_REPOSI-TORY_ITEMS	10,000 items	NFR-KR-1	Team estimate: an upper bound on the documents, models, and test records the club could accumulate over many years.
BACKUP_INTER-VAL	7 days	NFR-KR-2	Team estimate: limits the loss to one week of contributions, which members could reasonably re-upload.

Constant	Value	Used in	Rationale / Reference
MAX_MODEL_LOAD_TIME	30 seconds	NFR-MA-1	Team estimate: STEP parsing and rendering is computationally expensive. This exceeds the 10 s limit for holding attention (Nielsen, 1993), so progress must be shown while loading. To be refined using Proof of Concept measurements.
DIMENSION_TOLERANCE	0.5 mm	NFR-MA-2	Team estimate: small relative to the thickness of a popsicle stick, so that extraction errors do not affect rule compliance results.
FAILURE_LOAD_TOLERANCE	25%	NFR-PA-1	Team estimate: the variability of hand-built popsicle stick bridges makes tighter predictions unrealistic for an initial model.
MAX_ANALYSIS_TIME	30 seconds	NFR-PA-2	Team estimate: analysis is requested occasionally rather than interactively, so a longer wait with a progress indicator is acceptable.
MAX_ROLE_SEARCH_TIME	2 minutes	NFR-KT-1	Team estimate: a new executive should be able to find their role record without assistance.
MIN_CONCURRENT_USERS	50 members	NFR-SYS-2	Club size: 4–6 registered teams plus executives and general members (Problem Statement).
MAX_ONBOARDING_TIME	1 week	NFR-SYS-4	Team estimate: allows a new software subteam member to contribute within their first week.

A.2 Changes from Template

Table 4 lists the changes made to the structure of the Meyer template and the reason for each.

Table 4: Changes from the Meyer template

Change	Reason
Added the Appendix book (A) after the Project book.	The Meyer template has only the Goals, Environment, System and Project books. The course rubric and SRS checklist require content that belongs in none of them (A.1, A.3, A.4).
Added A.1 Symbolic Constants.	The SRS checklist requires the values of symbolic constants to be given in a table in the Appendix.
Added A.2 Changes from Template.	The SRS checklist requires changes made to the template to be described in the document.
Added A.3 Generative AI Use Disclosure.	Required by the course rubric.
Added A.4 Reflection.	Required by the course rubric, using the questions from the course's SRS reflection template.

A.3 Generative AI Use Disclosure

A.3.1 AI Tools Used

Comment: *This section should not be empty!*

A.3.2 How and Where Used

Comment: *This section should not be empty!*

A.3.3 Verification of AI-Generated Content

Comment: *This section should not be empty!*

A.4 Reflection

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

A.4.1 Team

Comment: *This section should not be empty!*

Questions 3 & 5 answered as a team:

1.

2.

A.4.2 Pedro Azevedo

Comment: *This section should not be empty!*

1.

2.

3.

4.

A.4.3 Mason Azzopardi

Comment: *This section should not be empty!*

1.

2.

3.

4.

A.4.4 Michael Roberts

Comment: *This section should not be empty!*

1.
2.
3.
4.

A.4.5 Jack Easton

Comment: *This section should not be empty!*

1.
2.
3.
4.